Supporting Information

Ion Asymmetry and Electrostatic Screening in Dense Solutions of Patchy Macroions

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This document contains six sections of supporting information for the main article. Figures and tables are numbered S1, S2, ... and sections S1, S2, ... Cross-references to the main article are typeset as, e.g., "main Fig. 3" or "main Sec. 3.1.2".

S1 Simulation details

S1.1 Simulation system

All systems consisted of $N_{\mathrm{mi}} = 30$ rigid macro-ions in a cubic simulation box of side length $L_{\mathrm{box}} = 149.64\,\text{Å}$, corresponding to a macro-ion volume fraction of $\phi_{\mathrm{p}} = 0.30$. The solvent was modeled as a dielectric continuum with the permittivity of water at $T = 298.15\,\mathrm{K}$, $\varepsilon_r = 78.36$.

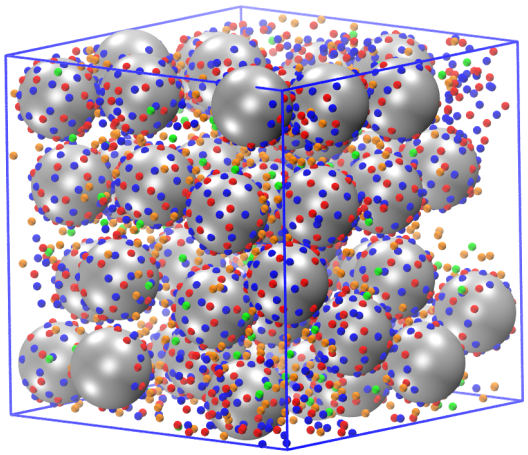


Figure S1: Simulation snapshot of the isotropic CPPM variant (P_0^0) in a periodic cubic box at volume fraction $\phi_\mathrm{p} = 0.30$ with NaCl ions at constant activity $a_\mathrm{NaCl} = 100\,\mathrm{mM}$. Color coding: gray = center particle, red = positive surface groups, blue = negative surface groups, orange = sodium ions, green = chloride ions.

Explicit-ion simulations. Monovalent sodium and chloride ions were included explicitly alongside the macro-ions. They were modeled as charged spherical beads of diameter $d_\mathrm{ion} = 4\,\text{Å}$ and elementary charge $\pm e$. Their chemical potential was controlled by grand-canonical Monte Carlo (GCMC) insertion/deletion of Na^+ and Cl^- pairs targeting a reference salt activity of $a_{Na^+} = a_{Cl^-} = 0.1\,\mathrm{M}$, with zero free-energy bias. For the three charged macro-ion variants the boxes initially accommodated $N_{Na^+} = 300$ sodium and $N_{Cl^-} = 60$ chloride ions on average. For the neutral variant, $N_{Na^+} = N_{Cl^-} = 123$ ions of each species were used to maintain electroneutrality. Those specific numbers were chosen based on preliminary simulations. A reference electrolyte simulation (bulk) was carried out in the same box without macro-ions.

Implicit-ion simulations. Implicit-ion simulations replaced the explicit ions with a Debye–Hückel screening term, parameterized by the Debye screening length κ^{-1} (main Eq. 4). Simulations were performed over the κ^{-1} ranges listed in main Table. 1. For each charged variant, an additional simulation was run at $\kappa_\mathrm{th}^{-1} = 9.61\,\text{Å}$ corresponding to an ionic strength of $I = 0.1\,\mathrm{M}$ [1].

Interaction potentials. Electrostatic interactions were evaluated with the Fanourgakis real-space, charge-neutral method [2] with a spherical cutoff $r_c = L_\mathrm{box}/2 = 74.82\,\text{Å}$. In the explicit-ion simulations the full Coulomb potential was used; in the implicit-ion simulations it was multiplied by the Debye–Hückel factor to give the Yukawa potential (main Eq. 4). Short-range repulsions between

all bead pairs were modeled with the Weeks–Chandler–Andersen (WCA) potential using Lorentz–Berthelot combining rules and well depth $\varepsilon = 2.479\,\mathrm{kJ\,mol^{-1}}$ ($= k_\mathrm{B}T$ at 298.15 K).

S1.2 Simulation protocol

Configurations were sampled with the Metropolis criterion over 4×10^4 macro-steps. Each macro-step combined: (i) rigid-body translation ($\Delta r_\mathrm{max} = 4.0\,\text{Å}$) and rotation ($\Delta\theta_\mathrm{max} = 0.4\,\mathrm{rad}$) trials for each macro-ion; (ii) preferential-sampling translations of the explicit ions biased towards macro-ion surfaces with weight $W(r) \propto (r + 5\,\text{Å})^{-2}$ and maximum displacement 14 Å; and (iii) GCMC insertion/deletion attempts for neutral $\mathrm{Na^+}$–$\mathrm{Cl^-}$ pairs. In the implicit-ion runs only moves of type (i) were applied. Three independent replicas were run for each variant and κ^{-1} value; all reported quantities are mean ± standard error across replicas.

S1.3 Analysis

Radial distribution function. CPPM–CPPM radial distribution functions $g(r)$ were computed from centers of mass every 10 MC steps (resolution 0.25 Å). The analytical Percus–Yevick (PY) hard-sphere $g(r)$ was computed as a theoretical reference. The PY structure factor $S(q)$ was evaluated on 50,000 uniformly spaced q-points up to $q_\mathrm{max} = 5.0\,\text{Å}$ using the exact Wertheim–Thiele solution [3, 4]. The pair distribution function was obtained via the sine transform

$$g(r) - 1 = \frac{1}{2\pi^2\rho} \int_0^{q_\mathrm{max}} q^2 \big[S(q) - 1\big] \frac{\sin(qr)}{qr} \mathrm{d}q, \tag{1}$$

with a Lorch window $W(q) = \sin(\pi q/q_\mathrm{max})/(\pi q/q_\mathrm{max})$ applied to suppress Gibbs truncation artifacts. The effective hard-sphere diameter d_eff was fitted to the neutral CPPM $g(r)$ by minimizing the root-mean-square deviation of $f(r) = g(r) \cdot r^2$; the effective volume fraction follows as $\phi_\mathrm{eff} = \phi_\mathrm{p}(d_\mathrm{eff}/d_\mathrm{cp})^3$.

In addition to CPPM–CPPM distribution functions, ion–CPPM pair distributions ($\mathrm{Na^+}$–NP, $\mathrm{Cl^-}$–PP, $\mathrm{Na^+}$–MP, $\mathrm{Cl^-}$–MP) were computed in the explicit-ion runs. A correction for the partial occlusion of each spherical shell by the macro-ion was applied for $r < d_\mathrm{eff} = d_\mathrm{cp} + d_\mathrm{sg} = 44\,\text{Å}$ by dividing by the accessible shell fraction $f_\mathrm{acc}(r) = (d_\mathrm{eff} + r)/(2\,d_\mathrm{eff})$. The $\mathrm{Na^+}$–$\mathrm{Cl^-}$ pair distribution was also computed, including the bulk NaCl reference.

Optimal screening-length selection. For each charged variant, the quality of each implicit-ion $g(r)$ was assessed by the peak absolute deviation $|\Delta g|_\mathrm{max}$ against the explicit-ion reference, restricted to $r \geq 40.0\,\text{Å}$ to focus on medium- to long-range structure. The κ^{-1} value minimizing $|\Delta g|_\mathrm{max}$ was identified as κ_*^{-1}; to improve precision beyond the discrete grid the $|\Delta g|_\mathrm{max}(\kappa^{-1})$ curve was interpolated with a cubic spline and minimized on 10,000 points spanning the simulated range.

Multipole energy decomposition. Pairwise CPPM–CPPM interaction energies were decomposed as a function of center-of-mass separation r using the Multipole Distribution analysis implemented in Faunus (resolution 0.25 Å, sampled every 10 steps). For each configuration, the exact energy u_{exact} (full bead–bead Coulomb sum) and the truncated multipole-expanded total $u_{\mathrm{tot}} = u_{ii}+u_{id}+u_{dd}+u_{iq}$ (main Eqs. 5–8) were recorded. The same analysis yielded orientational correlation functions as a function of r: (a) dipole–dipole correlation $\langle\hat{\mu}_i \cdot \hat{\mu}_j\rangle$; (b) longitudinal dipole projection $\langle(\hat{\mu}_i \cdot \hat{r}_{ij})(\hat{\mu}_j \cdot \hat{r}_{ij})\rangle$; (c) quadrupole–quadrupole correlation $\langle\hat{\mathbf{Q}}_i{:}\hat{\mathbf{Q}}_j\rangle$; and (d) nematic order parameter $\langle P_2(\cos\theta_{ij})\rangle = \langle(3\cos^2\theta_{ij}-1)/2\rangle$.

Spatial distribution functions. Three-dimensional density maps of surface beads (NP, PP) and explicit ions (Na^+, Cl^-) around the macro-ion reference frame were accumulated every 10 steps at 1.5 Å resolution with 8 Å padding and projected onto the unit sphere as Mollweide equal-area maps.

Osmotic pressure and activity coefficient. The total osmotic pressure $p^{\mathrm{osm}}_{\mathrm{total}} = p_{\mathrm{ideal}} + p_{\mathrm{excess}}$ comprises ideal and excess contributions. In the explicit-ion simulations the ideal part follows from the mean number densities of macro-ions, Na^+, and Cl^-. In the implicit-ion runs, the ideal contribution reduces to the macro-ion number density alone. The mean activity coefficient is estimated as $\gamma_\pm = \sqrt{\gamma_+\gamma_-} = \sqrt{a_+/c_+ \cdot a_-/c_-}$. The excess contribution was sampled by virtual isotropic volume perturbations ($\delta V = 0.1\,\text{Å}^3$, block size 1000 steps). The total osmotic pressure increment is computed as $\Delta p^{\mathrm{osm}}_{\mathrm{total}} = p^{\mathrm{osm}}_{\mathrm{total}} - p^{\mathrm{osm,bulk}}_{\mathrm{total}}$.

S2 Orientational correlation analysis

The isotropic variant P_0^0 shows no significant orientational order at any separation (|correlation| < 0.02 for all four metrics), confirming that packing effects — not electrostatic anisotropy — control the structure of this system, consistent with the hard-sphere-like $g(r)$ discussed in main Sec. 3.1.2.

The dipolar P_8^1 exhibits a clear positive dipole–dipole correlation ($\langle\hat{\mu}_i \cdot \hat{\mu}_j\rangle \approx 0.17$ at $r \approx 44$ Å) and an equally significant positive longitudinal projection (≈ 0.19), indicating that neighboring dipolar macro-ions tend to adopt a head-to-tail configuration — the energetically preferred arrangement for two interacting dipoles at close range. The negative nematic order parameter ($\langle P_2\rangle \approx -0.06$) is consistent with a broad angular distribution where the head-to-tail motif dominates but perpendicular configurations also occur. The quadrupole–quadrupole correlation of P_8^1 is small and negative (≈ -0.085); the primary orientational signature is dipolar. All correlations in P_8^1 decay to noise by $r \approx 52$ Å, indicating that orientational memory is confined to the contact shell.

The quadrupolar P_8^2 shows negligible dipole–dipole correlation (< 0.01) but a large negative quadrupole–quadrupole correlation ($\langle\hat{\mathbf{Q}}_i{:}\hat{\mathbf{Q}}_j\rangle \approx -0.19$ at $r \approx 44$ Å), indicating dominant anti-aligned quadrupolar ordering at contact: the

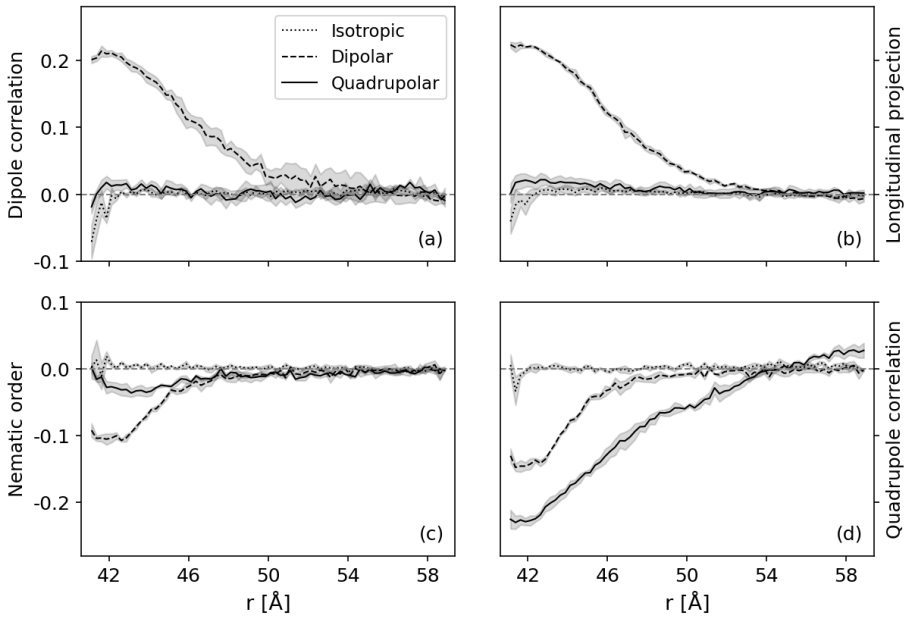


Figure S2: Distance-dependent orientational pair correlations between macroions for the three charged variants P_0^0 (dotted), P_8^1 (dashed), and P_8^2 (solid) as a function of center-of-mass separation r. (a) Dipole–dipole correlation $\langle \hat{\mu}_i \cdot \hat{\mu}_j \rangle$; (b) longitudinal dipole projection $\langle (\hat{\mu}_i \cdot \hat{r}_{ij})(\hat{\mu}_j \cdot \hat{r}_{ij}) \rangle$; (c) nematic order parameter $\langle P_2(\cos\theta_{ij}) \rangle = (3\cos^2\theta_{ij} - 1)/2$; (d) quadrupole–quadrupole correlation $\langle \hat{\mathbf{Q}}_i : \hat{\mathbf{Q}}_j \rangle$. Shaded bands: mean ± SEM across replicas. Horizontal gray line: uncorrelated reference ($= 0$).

two positive patches of one macro-ion face the regions of highest negative charge density on its neighbor. The quadrupole correlation decays more slowly than the dipole correlations in P_8^1, persisting to $r \approx 55\,\text{Å}$, consistent with the slightly broader first-shell peak in main Fig. 3.

S3 Multipole energy decomposition from explicit-ion simulations

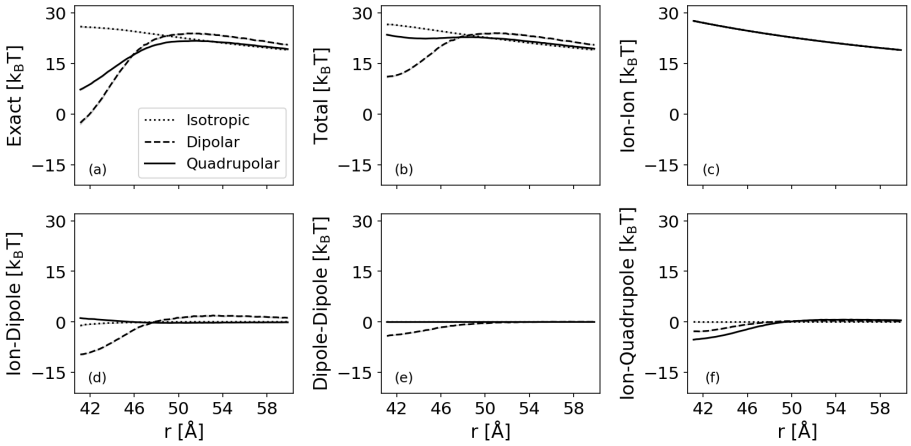


Figure S3: Distance-dependent pairwise energy contributions between macro-ions from explicit-ion simulations, for the three charged variants P_0^0 (dotted), P_8^1 (dashed), and P_8^2 (solid) as a function of center-of-mass separation r. (a) Exact total energy u_{exact}; (b) multipole-expanded total u_{tot}; (c) ion–ion u_{ii}; (d) ion–dipole u_{id}; (e) dipole–dipole u_{dd}; (f) ion–quadrupole u_{iq}. Shaded bands: mean ± SEM across replicas.

Fig. S3 compares the exact total pairwise energy u_{exact} (full bead–bead Coulomb sum) against the truncated multipole-expanded total $u_{\text{tot}} = u_{ii} + u_{id} + u_{dd} + u_{iq}$ (defined in main Eqs. 5–8) as a function of pair separation.

For the isotropic P_0^0 variant the two quantities agree to within 1.8% at contact ($r \approx 43\,\text{Å}$), and the fractional deviation falls below 0.1% by $r \approx 48\,\text{Å}$. This rapid convergence is expected: a nearly uniform charge distribution keeps the higher-order multipole moments small, so the ion–ion term u_{ii} dominates at all relevant separations.

For the patchy variants the behavior is qualitatively different. The dipolar P_8^1 shows a 275% overestimate at contact ($u_{\text{tot}} \approx 12.6\,k_{\text{B}}T$ vs. $u_{\text{exact}} \approx 3.3\,k_{\text{B}}T$): in favorable head-to-tail configurations the positive patch of one macro-ion faces the negative patch of its neighbor, producing a large attractive higher-order contribution (dipole–quadrupole, quadrupole–quadrupole) that the truncated expansion completely misses. The error drops to 3.7% at $r \approx 48\,\text{Å}$ and below

0.2% at $r \approx 52\,\text{Å}$. The quadrupolar P_8^2 shows a 110% overestimate at contact ($u_{\text{tot}} \approx 22.6\,k_{\text{B}}T$ vs. $u_{\text{exact}} \approx 10.8\,k_{\text{B}}T$) and converges more slowly: a 10.6% error persists at $r = 48\,\text{Å}$ and $< 1\%$ is not achieved until $r \approx 60\,\text{Å}$. The slower convergence for P_8^2 reflects the more spatially extended patch arrangement — two patches separated by the particle diameter create a richer higher-order multipole landscape that requires larger separations to enter the asymptotic regime.

These results demonstrate that the truncated multipole expansion is reliable only beyond $r \approx 52$–$60\,\text{Å}$ (depending on anisotropy), well outside the 44 Å contact distance. Within the first coordination shell — where structural and thermodynamic properties are primarily determined — higher-order multipole terms make quantitatively significant contributions for the patchy variants, underscoring the importance of the full bead–bead Coulomb treatment in explicit-ion simulations.

S4 Mollweide projections of the ion distribution on the CPPM surface

The Mollweide equal-area projections in Figs. S4–S5 present the same ion-distribution data discussed in main Sec. 3.1.2 in a format that resolves the full angular structure without perspective distortion. For the dipolar P_8^1 (Fig. S4, top), the chloride enrichment is confined to a single compact lobe coinciding with the positive patch (subpanel a), while sodium covers the rest of the sphere uniformly (subpanel b). The positive-group isosurface of neighboring macro-ions (subpanel d) appears in the equatorial hemisphere opposite the patch, consistent with the head-to-tail pairing identified in Sec. S2. The quadrupolar P_8^2 (Fig. S4, bottom) shows two symmetric chloride lobes above the two positive patches and four well-localized positive-group spots, reflecting the two-fold symmetry of the anti-aligned quadrupolar ordering. For the isotropic P_0^0 (Fig. S5, top) both ion species are nearly uniform across the sphere, with sodium enrichment everywhere and a diffuse, patternless chloride distribution. The neutral P^{n} (Fig. S5, bottom) shows both species uniformly and weakly distributed, confirming the absence of any electrostatically driven selectivity.

S5 Debye length scan

Fig. S6 shows the full dependence of $g(r)$ on κ^{-1} for the three charged variants. For the dipolar P_8^1 and quadrupolar P_8^2, the peak height tracks monotonically with κ^{-1}: stronger screening (smaller κ^{-1}) increases the peak because the reduced long-range repulsion allows closer approach. For the isotropic P_0^0, the peak height also increases with stronger screening, but every implicit-ion $g(r)$ systematically overestimates the explicit-ion reference across the entire scanned range ($\kappa^{-1} = 10$–$38\,\text{Å}$), confirming the fundamental limitation of the implicit-ion model for this variant discussed in main Sec. 3.2.

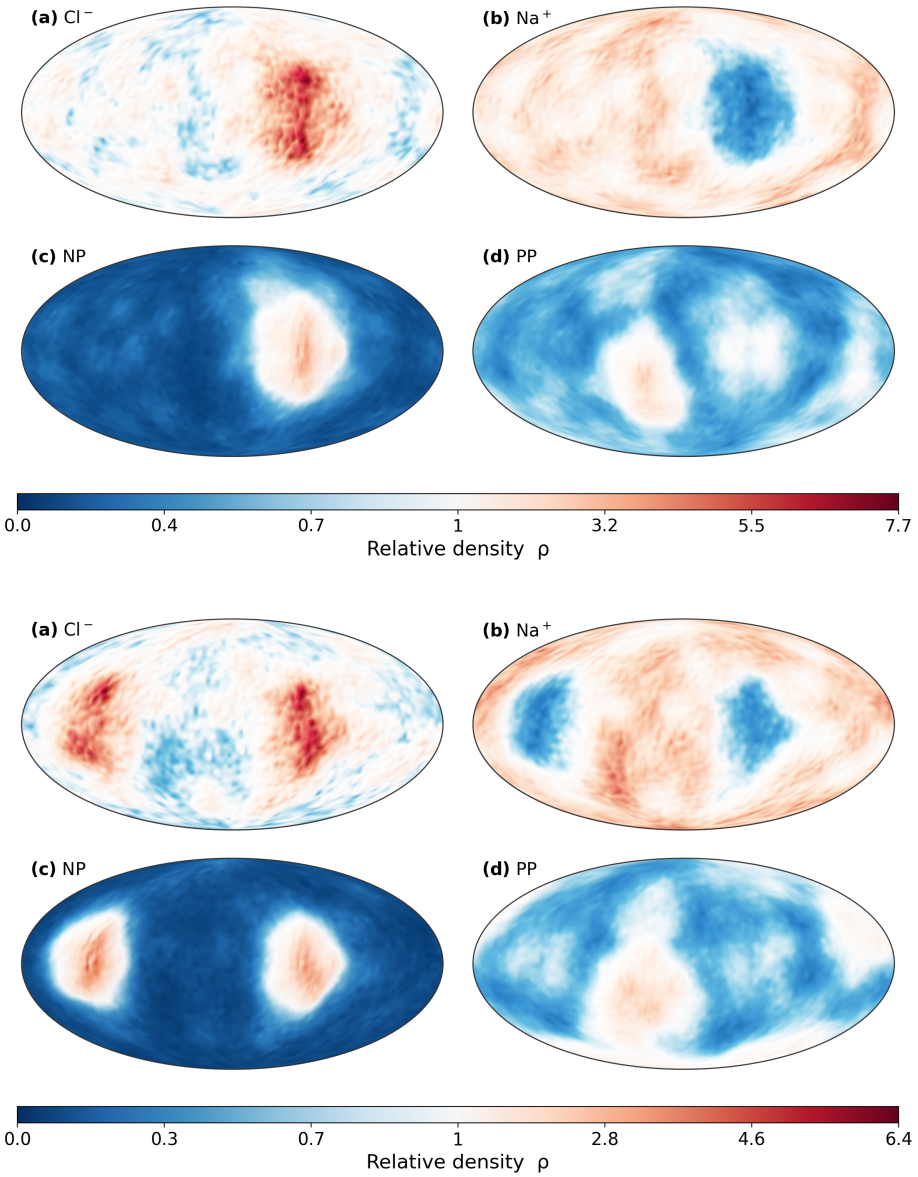


Figure S4: Spatial distribution of salt ions and surface groups of neighboring macro-ions around each CPPM variant, projected onto a Mollweide equal-area map of the unit sphere. Local density $\rho/\rho_{\mathrm{bulk}}$ sampled on a spherical shell at fixed center-of-mass separation: $\rho = 1$ (white) = bulk; $\rho > 1$ = enrichment; $\rho < 1$ = depletion. Subpanels: (a) chloride; (b) sodium; (c) NP surface groups; (d) PP surface groups. The color scale is shared within each variant. (Top) Dipolar P_8^1 at $r = 25\,\text{Å}$. (Bottom) Quadrupolar P_8^2 at $r = 25\,\text{Å}$.

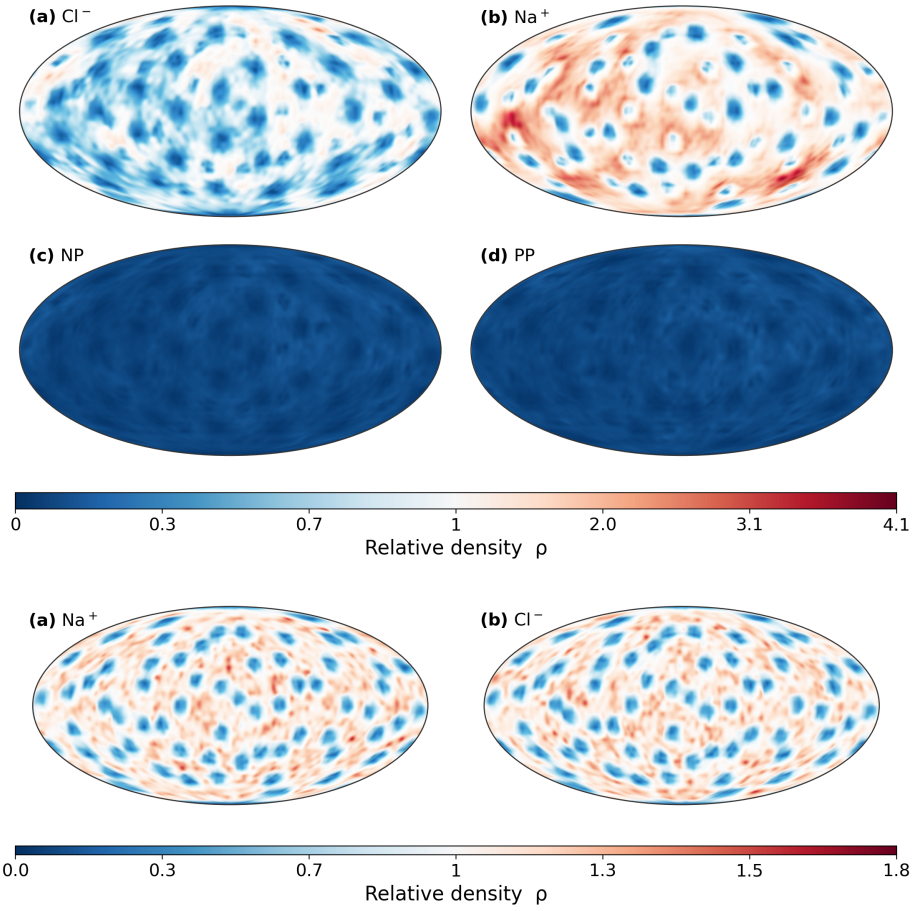


Figure S5: Same as Fig. S4 for the remaining variants. (Top) Isotropic P_0^0 at $r = 25\,\text{Å}$. (Bottom) Neutral P^{n} at $r = 23\,\text{Å}$; subpanels (c) and (d) are omitted because surface groups carry no charge.

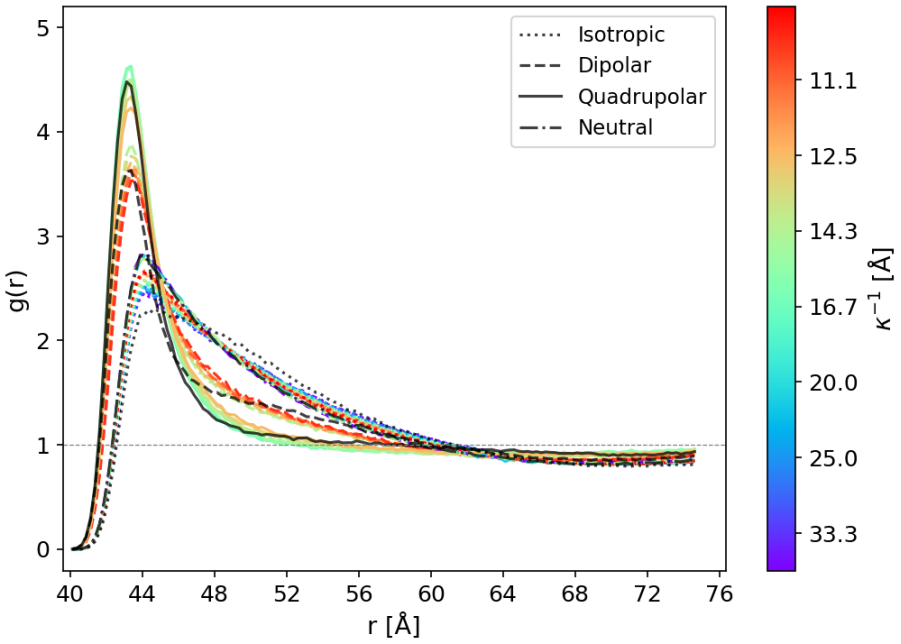


Figure S6: Center-of-mass $g(r)$ for the three charged variants over the full κ^{-1} scan from implicit-ion simulations. P_0^0 (dotted), P_8^1 (dashed), P_8^2 (solid). Line color encodes κ^{-1} (rainbow bar: violet = weak screening, large κ^{-1}; red = strong screening, small κ^{-1}). Black lines: explicit-ion reference. Shaded bands: mean ± SEM.

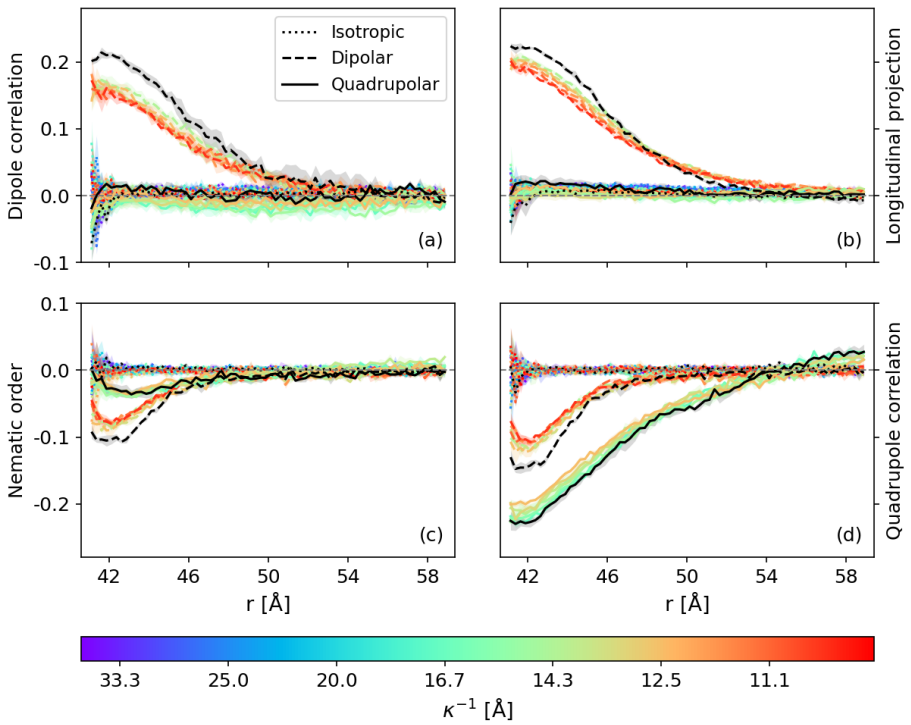


Figure S7: Orientational pair correlations for the three charged variants over the full κ^{-1} scan. (a) $\langle \hat{\mu}_i \cdot \hat{\mu}_j \rangle$; (b) $\langle (\hat{\mu}_i \cdot \hat{r}_{ij})(\hat{\mu}_j \cdot \hat{r}_{ij}) \rangle$; (c) $\langle P_2(\cos\theta_{ij}) \rangle$; (d) $\langle \hat{\mathbf{Q}}_i : \hat{\mathbf{Q}}_j \rangle$. Rainbow color bar encodes κ^{-1}; black lines: explicit-ion reference; shaded bands: mean $\pm$ SEM.

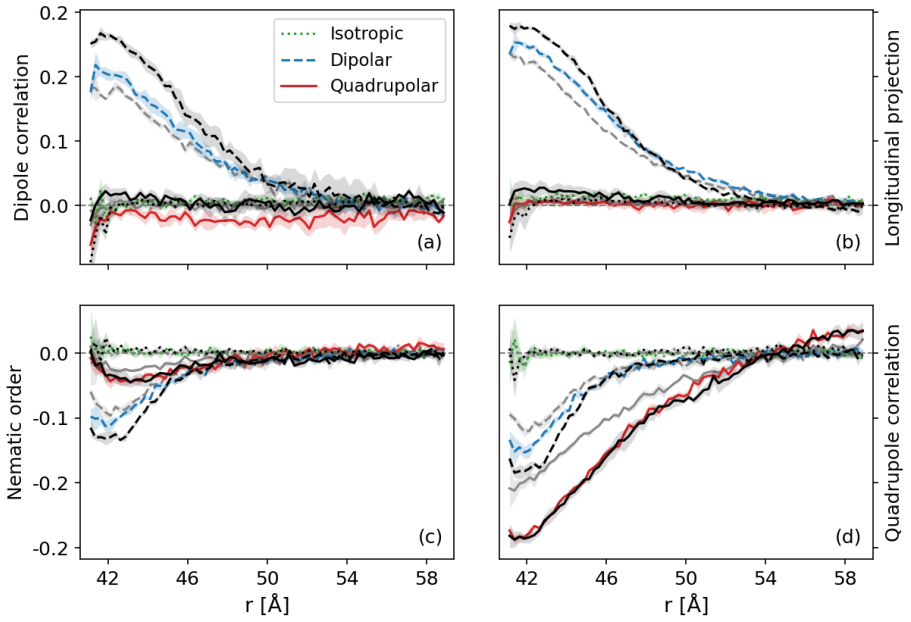


Figure S8: Orientational pair correlations at the best-fit screening length κ_*^{-1} (colored lines, one per variant) and at the theoretically predicted $\kappa_{\mathrm{th}}^{-1}$ (gray lines), compared with the explicit-ion reference (black lines). Panel layout as in Fig. S7. Shaded bands: mean ± SEM.

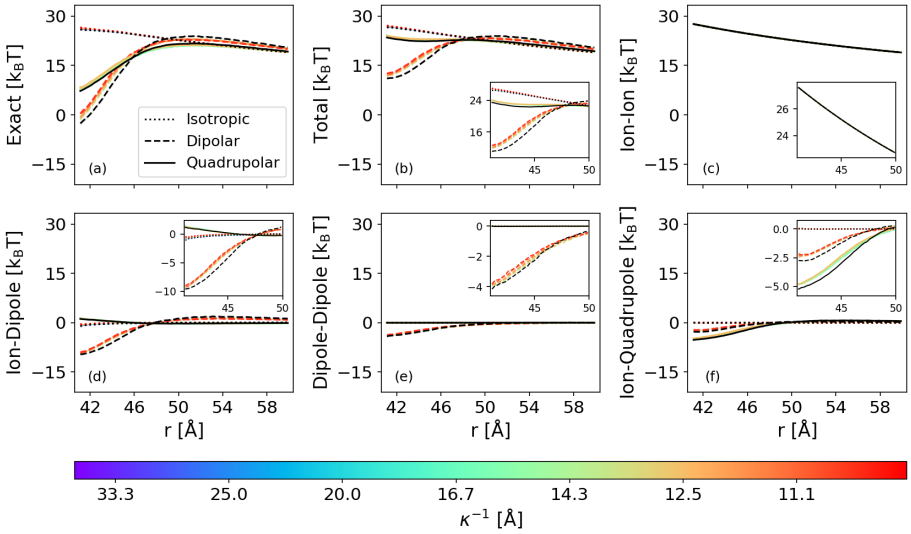


Figure S9: Distance-dependent pairwise energy contributions over the full κ^{-1} scan. Panel layout as in Fig. S3 with an individual energy axis per panel. Rainbow color bar encodes κ^{-1}; black lines: explicit-ion reference; shaded bands: mean ± SEM. Insets: zoomed view of the contact region ($r \leq 50$ Å).

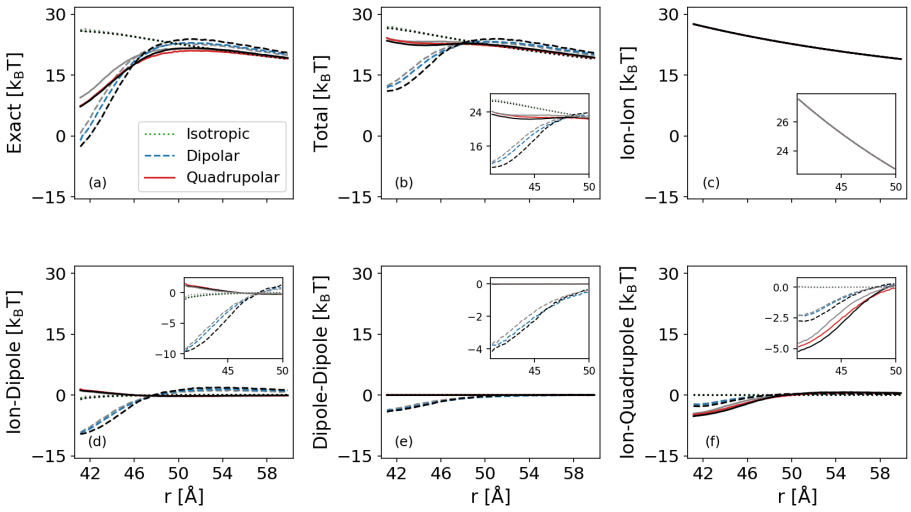


Figure S10: Pairwise energy contributions at κ_*^{-1} (colored) and κ_{th}^{-1} (gray) vs. the explicit-ion reference (black). Panel layout as in Fig. S3. Shaded bands: mean ± SEM. Insets: zoomed view of the contact region ($r \leq 50$ Å).

Figures S7–S8 extend the comparison to orientational correlations. The evolution of the dipole–dipole and quadrupole–quadrupole correlations with κ^{-1} closely tracks the $g(r)$ peak: stronger screening amplifies orientational order, and the optimal κ_*^{-1} recovers not only the correct $g(r)$ peak height but also the correct orientational order for P_8^1 and P_8^2 (main Sec. 3.2.2).

Figures S9–S10 compare the pairwise multipole energy components across the κ^{-1} scan and at κ_*^{-1}. The implicit-ion ion–ion term u_{ii} (panel c in both figures) varies strongly with κ^{-1} and converges to the explicit-ion reference at the best-fit screening length for P_8^1 and P_8^2, while remaining discrepant for P_0^0 at all values tested.

S6 Energy convergence

Figures S11–S15 show the running-mean total energy $U/k_{\mathrm{B}}T$ as a function of Monte Carlo sweeps for all simulations reported in the main article. The stabilization of the cumulative mean (thick lines) confirms convergence for all systems and replicas.

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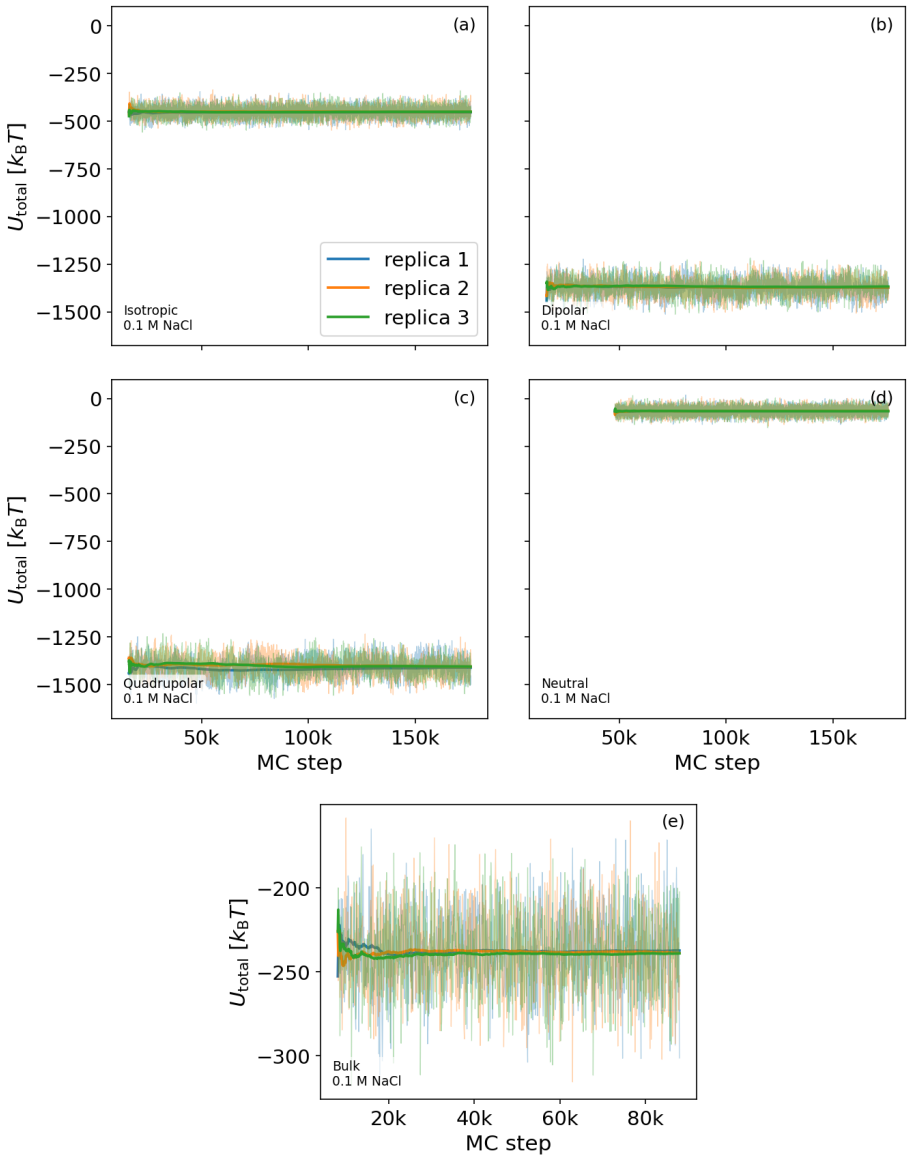


Figure S11: Total energy vs. MC sweeps for the explicit-ion simulations: (a) P_0^0, (b) P_8^1, (c) P_8^2, (d) P^{n}, (e) bulk NaCl. Thin lines: raw trace; thick lines: running cumulative mean. Colors distinguish replicas (blue, orange, green). Panels (a)–(d) share a common energy axis; (e) has its own scale.

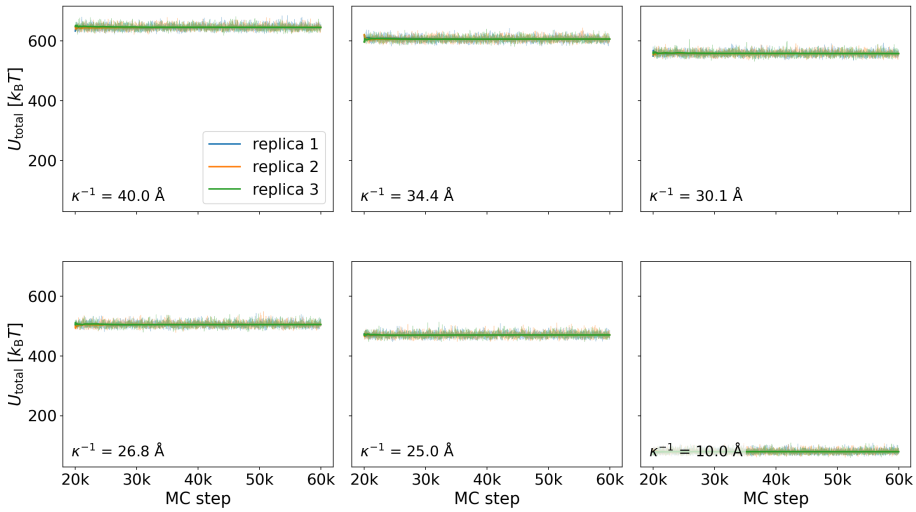


Figure S12: Total energy vs. MC sweeps for the implicit-ion simulations of P_0^0. Each panel shows one value of κ^{-1} (indicated in the lower-left corner). Thin lines: raw trace; thick lines: running cumulative mean. Colors distinguish replicas. All panels share a common energy axis.

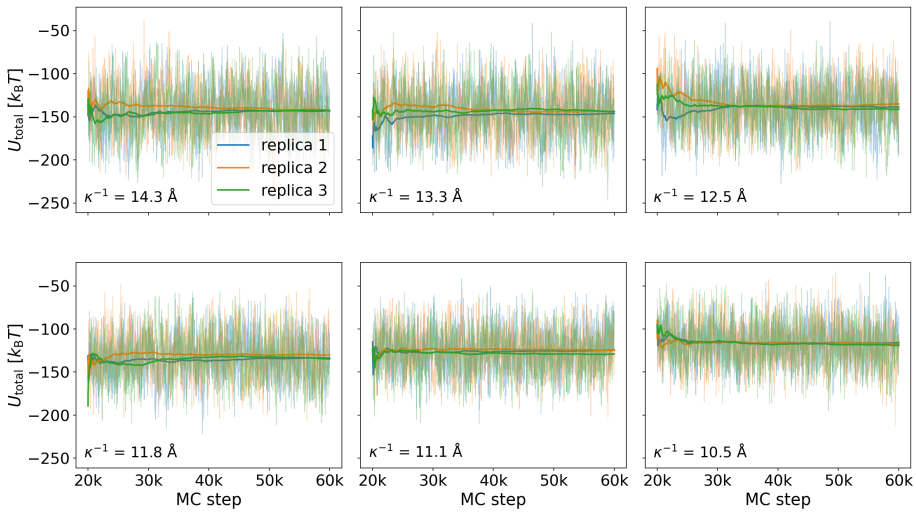


Figure S13: As Fig. S12 for P_8^1.

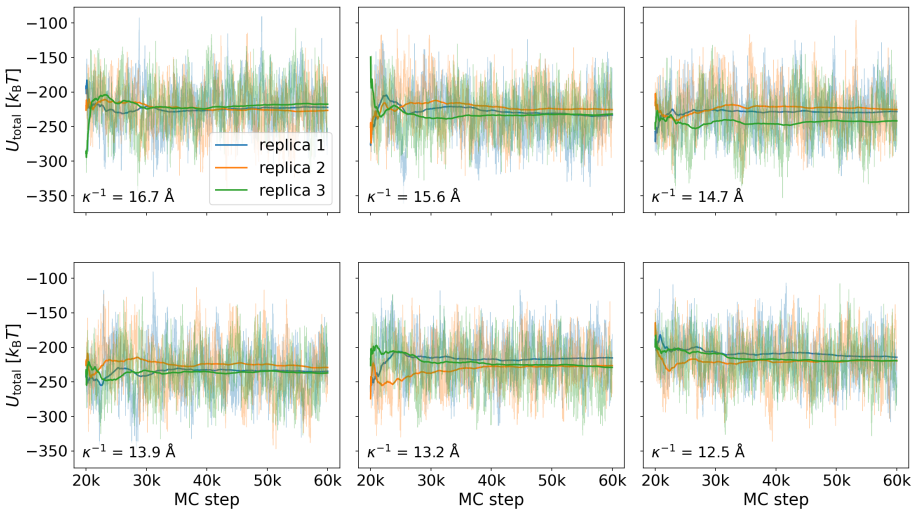


Figure S14: As Fig. S12 for P_8^2.

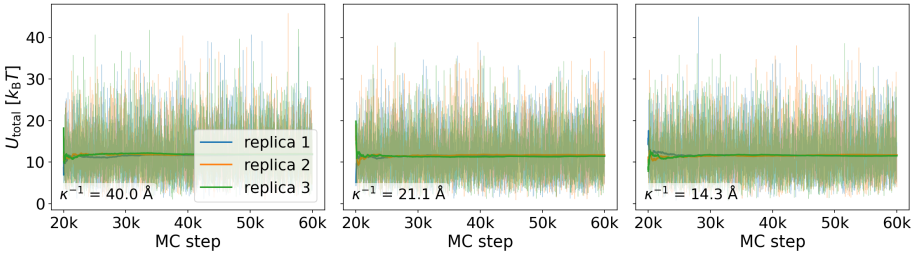


Figure S15: As Fig. S12 for P^{n}.